

Prithwjit Ghosh

Data Scientist | Machine Learning | GenAI

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GitHub | Portfolio | LinkedIn

Data Scientist Specialist, Accenture, with **3+** years' experience and an M.Sc. in Statistics from **IIT Kanpur (AIR 7)**, building production ML systems and forecasting pipelines (**95–99%** accuracy, 48 months out) across QSR, finance, and utilities. Deployed **agentic GenAI** platforms across **70,000+** enterprise contracts for **7 clients**, mentoring **3–4** junior data scientists.

Professional Experience

Data Scientist Specialist | *Accenture Global Technology*
Bengaluru, Karnataka, India

July 2023 – Present

● Diamond Projects

• **Sales and Guest-count Forecasting** | *Global Quick Service Restaurant (QSR) Brand*

- Architected and productionized an end-to-end ensemble forecasting system using Prophet, Theta, MSTL, LightGBM, and Naive models across **6** countries to accurately predict sales and guest counts up to **48 months** into the future
- Designed and owned a custom horizon-aware evaluation framework spanning **450+** forecasting models and a weighted ensemble balancing strategy, ultimately achieving **97-99%** accuracy at **24 months** and **95-96%** accuracy at **48 months**
- Established a production-grade **MLOps** pipeline with **Docker** containers, GitHub Actions **CI/CD**, and SonarQube/Snyk **security** scanning, running fully automated monthly forecasts through Airflow **DAGs** orchestrated on AWS **Fargate** infrastructure

• **Intelligent Collections 3.0** | *Water Treatment Brand*

- Developed customer-level late payment prediction models using **XGBoost**, selecting the **top 50** optimal features from **1000+** candidates and applying **quantile calibration** to reach **84-90% AUC** across a **10,000+** monthly customer base
- Formulated short-term accounts receivable forecasting models using both statistical and ML/DL methods, consistently reaching **95-98%** accuracy and partnering with finance leadership to deliver results through interactive **Power BI** dashboards
- Led deployment of calibrated risk scores integrated with accounts receivable forecasts into collections workflows, reducing overdue amounts by **38%**, lowering open accounts receivable by **15%**, and increasing collections by **12%** over two years

• **Financial Forecast Intelligence** | *Accenture Technology*

- Spearheaded development of a natural language decision intelligence platform over contract profit and loss data, answering finance managers' questions on margin and cost drivers across **70,000+** active contracts
- Orchestrated a multi-agent system using LangChain, LangGraph, Langfuse and **Gemini**, with a typed **intent SQL compiler**, **8** financial agents, and a **53-node** knowledge graph, replacing LLM routing with deterministic **graph** traversal
- Contributed to **Azure** migration consolidating a cross-cloud stack onto Container Apps, ACR, Key Vault, and Cache for Redis, integrating Azure OpenAI with managed identity and automating deployment via Azure DevOps CI/CD

● Short Projects

• **Machine Failure Alert System** | *Global QSR Brand*

- Designed a shake machine failure alert system with a statistical and XGBoost/Decision Tree ensemble (**93%** accuracy, **92%** recall) vs. 16% baseline, extending failure lead time from **3-5** to **30 minutes** for early intervention
- Architected the full production pipeline end-to-end using **Kafka** and S3 for ingestion, AWS-based training with daily inference on **edge servers**, Prometheus/Grafana for monitoring, and **NATS** for real-time alert delivery

• **Cash Flow Forecasting** | *Power Utility Company*

- Built and productionized end-to-end **B2C** cash flow forecasting for **6M+** customers using advanced ML modeling, improving **Cash-In** accuracy from 70% to **98%** and **Cash-Out** from 64% to **93%**, with insights surfaced through interactive dashboards

Skills

- **Languages:** Python, SQL, R
- **BI/ Analytics Tools:** Power BI, Streamlit, MS Excel, MS PowerPoint
- **Cloud:** AWS (Fargate, EMR, EC2, S3, CloudWatch, SageMaker), Azure (VM, ACA, AKS, Blob Storage, Azure ML)
- **GenAI/LLM:** LangChain, LangGraph, RAG, AI Agents, Agentic AI, OpenAI, Claude, Ollama, Llama, Mistral
- **ML/DL Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Prophet, TensorFlow, Keras, NLTK, spaCy
- **Deployment:** Docker, Airflow, Astronomer, GitHub Actions, Git
- **Microservices/ API:** FastAPI, Redis, Prometheus, Grafana, Kafka
- **Domains:** Predictive Modeling, Time Series Forecasting, Unsupervised Learning, NLP, Statistics, A/B Testing, MLOps

Education

- **Master of Science in Statistics**, Indian Institute of Technology Kanpur (IITK) Aug'21 - Jul'23
CGPA: 8.9 | JAM Rank: 7 (AIR) | Department Rank: 6
- **Bachelor of Science in Statistics**, Bidhannagar College Jun'18 - Jul'21
CGPA: 9.99 | Department Rank: 1